

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: Tsang Ho Nam

Student ID: 22077009D

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf_

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0 (toward R1)

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? YES

Does R2 show up as an OSPF neighbor on R3? NO

What does this information tell you?

R2's S0/0/0 interface was made non-passive, so R2 can only form an OSPF adjacency with R1, not with R3. R2's S0/0/1 is still passive, preventing Hello packets from being sent to R3. Therefore, R3 must reach the 192.168.2.0/24 network indirectly through R1 (cost = 64 + 64 + 1 = 129) instead of directly through R2.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1 (direct link to R2)

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

64+1=65

Is R2 listed as an OSPF neighbor to R3? YES

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

It causes all links with speeds above 100 Mbps to have the same OSPF cost of 1. It means OSPF cannot distinguish between different speed links.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

Cost to 192.168.3.0/24 = $781 + 1 = 782$. Cost to 192.168.23.0/30 = $781 + 64 = 845$.

R1's serial interfaces have a cost of $100,000 \div 128 = 781$ because the bandwidth was set to 128 kb/s, while R3's serial interfaces still use the default bandwidth of 1544 kb/s, giving them a cost of 64.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

After issuing **bandwidth 128** on all serial interfaces in the topology, every serial interface now has a cost of $100,000 \div 128 = 781$. The route from R1 to 192.168.23.0/30 traverses two serial links: R1 S0/0/1 + R3 S0/0/1 = $781 + 781 = 1562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The **ip ospf cost 1565** command was applied to R1's S0/0/1 interface, making the direct path to R3 cost $1565 + 1$ (R3 G0/0) = 1566. The path through R2 via S0/0/0 costs 781 (R1 S0/0/0) + 781 (R2 S0/0/1) + 1 (R3 G0/0) = 1563, which is lower than 1566. Since OSPF always selects the lowest-cost path, the route to 192.168.3.0/24 now goes through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

The OSPF router ID uniquely identifies each router in the OSPF domain. It is used in the DR/BDR election process and to identify the originator of LSAs. If router IDs are not deliberately controlled, they are derived from interface IP addresses, which may change if an interface goes down, potentially causing OSPF instability and routing disruptions.

2. Why is the DR/BDR election process not a concern in this lab?

The DR/BDR election only takes place on multi-access network segments. In this lab, all router-to-router links are point-to-point serial connections that do not require a DR/BDR election. The only broadcast segments are the G0/0 LANs, but each LAN has only one router, so the election has no practical impact.

3. Why would you want to set an OSPF interface to passive?

Setting an interface to passive prevents OSPF from sending Hello packets on that interface. This reduces unnecessary routing protocol traffic on segments where no OSPF neighbors exist (such as LAN segments with only end hosts) and improves security by preventing unauthorized routers from forming OSPF adjacencies on those interfaces.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The use of multiple areas. OSPF can divide a large network into separate areas, which reduces the size of the link-state database on each router and limits the scope of SPF recalculations, enabling the network to scale to a very large size.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route will be used. When a router learns routes to the same destination from multiple routing protocols, it selects the route with the lowest administrative distance (AD). EIGRP has an AD of 90, OSPF has an AD of 110, and RIP has an AD of 120. Since EIGRP's AD is the lowest, its route is installed in the routing table regardless of the individual metric values.